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RECENT INFORMATION ON SALMON STOCKS IN BRITISH COLUMBIA

Compiled by W.E. Ricker and J.I. Manzer
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(for the use of the Scientific Subcommittee of the Ad Hoc Committee on Abstention of the International North Pacific Fisheries Commission at its 1967 meeting.)

At the 1966 meeting of the Scientific Subcommittee of the Ad Hoc Committee on Abstention of the INPFC, Japan asked for "information on recent changes in the fisheries and in the conditions of the stocks of salmon, halibut and herring that are still under abstention" (INPFC Proceedings, 13th Annual Meeting, p. 264). In an explanatory statement it was indicated that they wished to have "not only a series of statistical figures but also explanations and evaluations of recent changes in fisheries and stocks" (Ibid., p. 267).

This paper has been prepared in answer to this request, in respect to British Columbia salmon. Tabular and graphical data on catches and escapements are presented through 1966, and interpretations of these data are given for the major groups of stocks.

The fact that British Columbia salmon met the three necessary qualifications for "abstention" was demonstrated in Canada's earlier submissions to the Commission. Recent trends indicate that all stocks are still in a condition where more intensive exploitation would not increase the maximum sustained yield. It is also still true that exploitation is everywhere regulated for the purpose of obtaining maximum productivity, and all important salmon stocks are, to varying degrees, the subject of scientific investigation.

SALMON STOCKS AND FISHERIES

Recent information on salmon landings is summarized in Tables 1 and 2. These data are shown graphically in Fig. 1-3, with landings plotted on a log scale. Regression lines shown were statistically fitted to logged values. The total salmon catch has continued to be variable, but there has been an average trend downward of about 2% per year from 1945 to 1966 (Fig. 1).

The breakdown by species shows that most of the decline has been in the chum salmon take, which began to fall off steeply in 1955 (Fig. 2). The average rate of decrease for 1945-1966 is nearly 8% per year. The other species show only minor and mostly non-significant overall trends, but much variability from year to year. Pinks were low in 1960 and 1965, but the Canadian catch reached an all-time high in 1962 (Fig. 2). Sockeye had a big year in 1958 and a poor one in 1963 (Fig. 3). Chinooks declined somewhat to 1962 but have increased again recently (Fig. 3). Cohoes had a poor year in 1960 but have increased rather steadily since, reaching new highs in 1965 and 1966 (Fig. 3).

The causes of these changes can be assessed best with respect to individual stocks or groups of stocks (runs), as presented later. However there are some considerations that apply rather generally. One of these is that there has been a substantial overall increase in fishing gear available to catch salmon, and it has become increasingly mobile and efficient. This has the advantage that it is possible to quickly concentrate enough gear to harvest any run that appears in exceptional abundance, even in a remote place. Its disadvantage is that there is much more than enough equipment and manpower to harvest ordinary-sized runs during most of the season, so that severe limitation of fishing operations is necessary to provide adequate escapements. This includes complete closures at certain times and places, and elsewhere the restriction of fishing to a portion of each week, sometimes to only 2 days or even one. Naturally there is great pressure from fishermen for the longest possible fishing time each year, since their immediate living is at stake and future rewards may often seem remote.

Another problem is that in many places it is difficult or impossible to harvest stocks selectively: that is, two or more species, or different stocks of the same species, are so intermingled even in inshore waters that they must be captured at the same time and in the same gear. One difficulty with this is that different stocks may migrate through the fishery at different rates and so stay in a fishing area for different lengths of time, so that a given number of nets will take a larger fraction of one stock than of another. Another difficulty, similar in effect, is that the stocks being fished together need not all have the same productive potential: that is, no single rate of utilization in a given year will provide optimum escapements for all of them. Different stocks vary in abundance in different ways from one year to the next, and certain stocks are inherently less productive* than others, so require a lower rate of cropping if their maximum yield is to be maintained. One step toward securing greatest possible flexibility in management has been the prohibition of net fishing in outside waters, where a great many stocks are mixed together and exploitation of individual stocks in accordance with their productivity is

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*The productivity of a stock, in the sense intended here, may be defined as the ratio of catch to total stock over a long period of years during which average catch is at its maximum.

practically impossible. This was agreed to by Canada and the United States in 1957.

Adjusting exploitation rates to the best level in each area each year has proved extremely difficult. This is reflected in the failure of the total catch and total salmon supply to increase since the 1940's, in spite of the fact that overall potential production is judged to be considerably greater than the present yield -- at least for sockeye, pinks and chums. Part of the trouble, but only part, is that forecasts of the size of individual runs in a given year are in many cases still not reliable enough to be really useful for management purposes. Techniques have been developed which measure the total stock as it develops. Rates of exploitation are provided by the same technique.

Variability in the ocean environment exists, and is reflected in different survival rates of successive year-classes of the various species. With important exceptions for particular species and areas, there is evidence that there have been mostly worse-than-average survival conditions for the pinks, chums and sockeye that returned during the past 8 years, approximately. Generally improved salmon supplies appeared in 1966 and, with the exception of pink salmon in the Central area, and coho salmon in general, have been followed by encouraging preliminary returns in 1967.

In comparing present yields with those made during early peak years of the fishery (between 1900 and 1920 in most cases), we must remember that the present sustainable yields will, in general, be less than the quantities actually taken in those early times. For this there are several possible reasons. One is a "fishing-up" effect: while spawning stocks were being reduced to the level that produces maximum sustained yield, there was (in some cases) a surplus to be harvested that made catches larger, temporarily, than the sustainable figure*. More important is inequality in productive potential or in lengths of exposure to fishing gear, as mentioned earlier. Calculations show that when stocks of similar timing but unequal productivity are fished in common, the less productive ones may decline greatly or even become extinct before the overall level of maximum sustained yield is reached (Ricker 1958a, Larkin and Ricker 1964b). Furthermore, this process can be quite protracted -- extending over 15 to 20 generations in extreme cases -- and while it is in progress catches remain greater than the sustainable level. Thus some runs may not, even yet, have had time to become completely adjusted to the "best" rate of utilization applicable to the whole group with which they are harvested.

A further possible factor may play an important role in the case of some sockeye stocks: a decline in productivity of their lake nurseries because of reduced fertilization by the carcasses of the spawners. The quantity of nutrients brought up to a nursery lake by a big spawning run is quite substantial -- up to at least 25 tons/km^2 (220 lb/acre) -- even in large lakes. Recent work on Lake Iliamna in Alaska (where the dominant year-class of 1965 brought 20 tons/km^2 of salmon into the lake) leads to the conclusion that "biogenic elements from decomposed salmon carcasses enrich the lake and enable it to support the progeny of the escapements in the peak cycle years." (Donaldson 1967a; see also 1967b). Thus the reduction in annual fertilization of a lake to half or a

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*This does not apply to stocks having dome-shaped reproduction curves such as "A" or "C" in Fig. 15; it does apply to smoothly convex curves such as "B" in Fig. 15 -- see Ricker 1958b, p. 247.

quarter of its potential level, which is unavoidable if a fishery is to exist, may in some cases directly reduce its productivity in the year concerned. It is also possible that reducing spawning stocks may lead to a cumulative impoverishment of some nutrients in lake basins where the rate of replacement of water is small. For either reason, the sockeye productive potential of some lakes may be less now than in the early years of the fishery.

During the past decade a stepped-up tempo of forestry operations and general industrialization, plus an increasing human population, have altered watersheds and diverted water supplies used by salmon. While ameliorative measures have usually been possible, some damage to salmon production is often unavoidable. So far all <u>major</u> watersheds have been preserved from hydro-electric encroachment or severe pollution, but at a cost to the country's economy which reflects the value placed on the salmon resource.

In order to increase salmon supplies, a number of artificial spawning channels with regulated flow have recently been constructed: at Jones Creek, Seton Creek and Weaver Creek in the Fraser system, at Fulton River in the Skeena system, and at Robertson Creek and Big Qualicum River on Vancouver Island. Most of these are only just beginning to come into production, and to some extent they are still experimental. Their total effect in increasing salmon supplies cannot be evaluated as yet: their size is such that, at best, they may increase the salmon catch of the province by about 2% over present levels. Additional channels are presently under construction and will increase this gain to about 4.5%. Other types of artificial assistance to reproduction are under study: two experimental hatcheries are in operation, at Lakelse in the Skeena system and on the upper Pitt River in the Fraser.

It should be noticed that, insofar as channels or hatcheries become substantially more efficient than natural reproduction, they immediately exacerbate the problem of selective harvesting. That is, _their_ stocks can support, and if possible should be given, a much more intensive rate of utilization than stocks that spawn under natural conditions; but any _general_ increase in rate of utilization of all fish would lead to a decline of the natural stocks affected.

Since the Canadian presentation on Abstention was first prepared, in 1957-58, a number of papers, bulletins and reports giving information on salmon stocks have become available. Many of these are listed in the references attached. In addition to material published anonymously in INPFC Bulletins and Annual Reports, the following papers deal directly with stock size and factors that may affect it: Aro (1961), Aro and Shepard (in press), Cooper (1965), Cooper and Henry (1962), Foskett (1958), Gilhousen (1960), Godfrey (1958a,b), Henry (1961), Hourston _et al_. (1965), Johnson (1958, 1965), Manzer and Shepard (1962), Margolis _et al_. (1966), Neave (1962), Palmer (1967), Parker (1962a,b, 1963, 1964, 1965), Ricker (1962b,c, 1964), Royal and Tully (1961), Ruggles (1965, 1966), Shepard (1962), Shepard _et al_. (1962), Shepard and Withler (1958), Shepard _et al_. (1964), Verhoeven and Davidoff (1962), Vernon (1958, 1966), Vernon _et al_. (1964), the Annual Reports of the International Pacific Salmon Fisheries Commission and of the Skeena Salmon Management Committee (cited under Withler and McDonald).

The ensuing sections discuss certain important stocks or groups of stocks in more detail. Abbreviations used are as follows:

BCCS - British Columbia Catch Statistics

CDFV - Canadian Department of Fisheries, Vancouver Region

FRBC - Fisheries Research Board of Canada

INPFC - International North Pacific Fisheries Commission

IPSFC - International Pacific Salmon Fisheries Commission

SSMC - Skeena Salmon Management Committee

The British Columbia Statistical Areas are illustrated in Fig. 4. The "Fraser River Convention Area", or "Convention Waters", in Canada include statistical areas 17-21, 23 (except Barkley Sound), 24 south of Latitude 49°, 28 east of Point Atkinson, all of 29, and the Fraser River; and in United States they include waters south to 48° Latitude and the coast of the Olympic Peninsula east to Point Wilson, thence to northwestern Whidbey Island and adjacent islands to the north.

FRASER RIVER SOCKEYE

Table 3 gives the recent history of the Fraser sockeye. The stocks as a whole are still substantially below most estimates of their maximum sustainable yield. Strenuous efforts to increase them are being made, but there are some special problems. The system contains a great many stocks, but two factors make considerable selectivity of utilization possible: differences in average times of arrival of the various stocks during a rather long fishing season, and the fact that some nursery areas can have precedence in making regulations because of their present contribution or large potential. However, there is much overlap in arrival times, and difficult decisions have to be made. Unpredictable changes in the deployment of fishing gear used by Canadian and United States fishermen, and the very short fishing period that is sometimes necessary in each week, tends to make precise regulation a touch-and-go matter.

In addition to differences in potential productivity of different stocks subject to common harvest, there are a number of areas of large capacity that are still in process of recovery from the Hell's Gate catastrophe of 1913-14, and for which special protection is desirable. Additional problems are created by occasional severe mortalities of certain stocks due to high river temperatures or other causes, and by the recent period of generally low marine survival rate.

Because of the 4-year "cycle" of many Fraser sockeye stocks, a picture of the trend for the river as a whole is best obtained by adding catches or escapements by sequences of 4 years. Tables 3 to 5 and Fig. 5 show the grouping that ends in 1966. The recovery phase of 1951-58 was followed by a decline in 1959-66; however spawning stocks have been maintained at the level of about 5 million (per 4 years) by close regulation.

The overall picture is dominated by the late Shuswap run (largely Adams River fish) in the 1902 line which as early as 1942 had reached an adult return level (about 11 million fish) which may be fairly close to the size providing maximum sustainable yield (Fig. 10). However on occasion this stock has been much more productive: it rose to a new high in 1954 (1950 brood year) and again in 1958 (1954 brood year, 19 million return), then slumped in the two subsequent generations. Escapements for this stock are shown in Fig. 7.

Other stocks, particularly the smaller ones, can be followed only in terms of their spawning escapements (Tables 4 and 5; Fig. 6-8), which show varied patterns. The Chilko, Quesnel and Seton-Anderson areas have exhibited a fairly steady increase; Nechako and the early Shuswap stocks have levelled off, not necessarily permanently. The Stuart and Bowron regions have lost part of the large gains made earlier. Downriver, most stocks have not changed importantly, but concern has been expressed about the Pitt and Lillooet regions. The latter historically supported the largest run below the canyon, and while Fig. 6 suggests a recovery during the past 8 years, most of this increase has been in the form of small jack sockeye.

Reproduction curves for various combinations of Fraser sockeye stocks have been plotted a number of times (Ricker 1954; Tanaka 1962a; Anon. 1962b; Doi 1962), with the understood limitation that the presence of many stocks in most of these combinations calls for special caution in their interpretation. New points can be added to these graphs from recent data (Fig. 9-12). Doi rightly points out that the variability of the returns is such that a number of different average reproduction curves can be fitted, all more or less plausible. The simple "Ricker-type" mathematical relationship used by Tanaka and Doi to fit some of their curves is a convenience, but there is no strong justification for using it on either theoretical or observational grounds. It has the disadvantage that, being based on the logarithms of observed stock sizes, it gives too little weight to years when the stock is large, and consequently it produces a trend line that is below the average level of return for any level of spawning. Accordingly, in drawing trend lines in Fig. 9-13 we have been guided by the positions of the points and by trends indicated by simple or running averages, rather than any mathematical relationship. In all cases the sums of deviations above and below the line shown are nearly equal -- though this condition of course does not of itself determine a unique position for the line.

The information from additional years added to Fig. 9-12 requires a general lowering of the outer part of the former trend line (whether freehand or computed) for the 1901 and 1902 cycles, and raising the outer part of the trend for the 1903 cycle (compare figures 9-12 in INPFC Bulletin 9, pp. 102-103). The change in the last named is associated with a shift in dominance of the Chilko stock from the 1904 to the 1903 cycle. This in turn points up the fact that insofar as there may be interaction between fish of the four cycles that results in dominance (Ricker 1950, Ward and Larkin 1964), reproduction curves for individual cycles are artificial and not very meanginful: in this situation only 4-year totals of a stock's productions and spawning escapements can give a measure of its performance, and be a guide to its potential.

Figure 13 shows the reproduction relationship for the system as a whole since 1915. Tanaka (1962a, p. 80) points out an apparent high productivity (per spawner) of the system during the period up to 1914 when the "big" years flourished, based on Rounsefell's (1949) estimates of spawning populations. Unfortunately Rounsefell's estimates of spawners were computed by an indirect method whose applicability to the pre-1917 data is very questionable, and his big-year escapements seem much too low in the light of the scattered direct observations that are available (see the discussion in Ricker 1950, p. 10). However it is only these years that give Tanaka's figure 10 a recognizable slope; hence no prediction of productivity per spawner can be made on this basis.

In any event, present regulation is not based on any overall computed relationship between spawner and returns, but rather on preliminary indications or forecasts of the abundance of the important runs in each individual year, and the estimated capacities of their spawning grounds and nursery lakes.

A cautiously optimistic outlook for the Fraser sockeye is still that it may be possible to bring the total yield up close to the level of harvest of 1896-1915 -- when an average of about 40 million sockeye per 4-year period was taken by Canada and the United States. This is about three times the average 4-year yield since 1951. Even more sanguine predictions are possible, based on the possibility that the dominant line was substantially underutilized during the early period. However, there is now a suggestion* that the high stock levels of the old big years depended partly on having <u>all</u> important upriver stocks in the same cycle, whereas present

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*This is from Henry's (1961, p. 72) observation that survival of <u>Chilko</u> smolts was much above expectation in two years of large <u>Shuswap</u> smolt runs -- suggesting that really great abundance of one stock increases the survival and productive potential of other stocks that may be associated with it at this stage of the life history.

dominant runs are in different cycles -- which is far more advantageous from the industry's point of view. Also there are the general conditions that tend to make sustainable productivity less than its early peak, which were discussed earlier. For these reasons the present average level of productivity of the Fraser system will be discoverable only by continued trials with larger spawning stocks, particularly in nursery lakes that seem greatly underutilized.

RIVERS INLET SOCKEYE

Yields of sockeye from Area 9, largely of the Rivers Inlet (Owikeno Lake) stocks, have averaged nearly 800,000 fish since 1951 (Table 6). Figure 14 gives a long-term picture. There is less indication of a long-term decline in average landings than in the Skeena area; in fact the trend through 1951-55 is barely perceptible and non-significant. However, adding the somewhat smaller yields of the past 10 years makes the trend formally significant.

Estimating escapement is difficult in some spawning streams of this area because of glacial silt in the water, so graphs of spawners and returns cannot be presented. Present levels of utilization are based on the empirical observation of spawners in rivers where they can be seen, and the general abundance of fish 4 and 5 years later.

General information on the Rivers Inlet stocks is presented by Foskett (1958) and Godfrey (1958a), while Ruggles (1965) has studied the fingerlings and smolts. Research on the biology, life history and population dynamics has recently been intensified. Stocks of Rivers Inlet and the adjacent Smith Inlet currently give the largest yields, per square mile of nursery lake surface, of any region in the province.

SKEENA RIVER SOCKEYE

Table 7 gives recent catches and escapements for the Skeena sockeye as a whole. For any analysis of success of reproduction in individual years a breakdown into age categories is necessary. One approach to this is in Table 1 of Appendix 2 of the Canadian presentation (Anon. 1962a, p. 31), which gives the quantity of sockeye caught on the Skeena and the "resultant return" from each spawning; that is, the catches were divided up according to their age composition and credited to the appropriate brood year. There is now available the more informative Table 8, which includes the escapement in the picture.

Figure 15 is the reproduction diagram with the new points added. The two new trend lines (B,C) shown for the outer part of the graph represent alternative possibilities for its extrapolation, the domed shape being closer to the available data. Tanaka (1962a,b) divided the Skeena spawner-return relationship into three productivity levels, using two different groupings, and fitted a theoretical curve to each. Most of these curves suggested that, at each level, maximum sustained yield would be obtained with a somewhat higher rate of utilization than at present. Reasons for largely rejecting this analysis have already been given (Anon. 1962b; Shepard _et al_. 1964). There is no objective evidence for three productivity levels, on either of the systems Tanaka uses -- in neither case do they represent continuous time sequences. Also, there is no evidence that the type of theoretical curve used is appropriate to these data. However this does not exclude the possibility of a decline in productivity; indeed some possible reasons for expecting one were presented earlier. In Fig. 15, the recent (circled) points to the right of 0.7 million spawners are all well below the average level of returns from spawnings of this size -- though not lower than some individual years in the past. Figure 16 compares average returns from average size categories of escapement during the recent period (1941-61) with those of earlier years. According to this system of averaging, the recent period produced less than the earlier one at all levels of spawning; however the dispersion of the points for individual years (Fig. 15) shows that it cannot yet be regarded as conclusive that a decline in productivity has occurred. That is, there is still a real possibility that the poor reproductive success of moderately large spawnings during the recent period may represent merely a random fluctuation.

Present management policy is testing both of these possible interpretations of the data by using both medium and moderately large spawning escapements, as occasion permits. Thus in a few years there will be reasonably firm objective evidence concerning the present maximum productivity level. According to Fig. 16, during the recent period a maximum sustained yield averaging 0.7 million sockeye would be obtained with spawning stocks anywhere between 0.6 and 0.9 million. During the earlier period the maximum surplus of about 1.0 million was produced by spawning stocks of about 1 million, though stocks as small as 0.6 million produced an average surplus not too much less (0.9 million).* The indicated maximum yield for both periods is obtained at rates of utilization between 45% and 55%. These figures agree with the values indicated by the two lower curves in Tanaka's (1962b)

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*However the averaging of all large spawnings in Fig. 16 conceals the possibility of a larger maximum surplus during the earlier period (about 1.4 million), from spawnings very close to 0.9 million (approximately Curve A in Fig. 15) -- this on the assumption that each brood is unaffected by the abundance or scarcity of neighbouring broods.

figure 5(I), but are considerably below the 76% indicated in his figure 5(II).

There is some possibility of interaction between adjacent year-classes on the Skeena, similar to what is given credit for producing dominant lines on the Fraser. Though the mixture of two major age-groups in the spawning stock makes an extreme development of "dominance" impossible on the Skeena, there have been two periods when one age-group predominated in the largest spawning stocks; this produced a 5-year cycle during 1914-24 and a 4-year cycle during 1936-52 (see INPFC Bull. 9, p. 30). If these escapements and returns be averaged by groups of 5 and 4 years, respectively, it appears that largest average surpluses have been produced when the average annual spawning was 0.6-0.7 million, while both smaller and larger average levels of spawning produced smaller surpluses. This is true of both the early and the recent period, though the indicated level of production is of course somewhat less in the later period.

One indication that the Skeena system's level of sustained yield may now be less than its early level of harvest is given by the present status of the smaller sockeye runs. These can be assessed only from estimates of their spawning escapements since 1944, shown in Table 9. Most of these lesser stocks have declined during the past 20 years, suggesting that fishing has been too intensive for their maximum steady yield. However they are so small that they cannot be specially managed in the common fishery with the Babine stocks, which in the 1940's accounted for about 75% of the annual Skeena escapement, and now about 90%.

PINK SALMON OF SOUTHERN BRITISH COLUMBIA AND NORTHERN WASHINGTON

Pink salmon of the southern Strait of Georgia, the Fraser River, the Strait of Juan de Fuca and northern Puget Sound were brought under international management by the International Pacific Salmon Fisheries Commission in 1957. Extensive surveys were made in 1959 and 1961 to determine the routes of migration and rates of utilization of the pinks in areas where these fish might be found alone or mixed with other runs (Vernon _et al_. 1964; Hourston _et al_. 1965); while more recent information is given in the IPSFC Annual Reports and annual reviews by CDFV (e.g. Palmer 1967).

From the middle Strait of Georgia southward appreciable quantities of pinks occur only in the odd-numbered years, the even-year line being virtually absent. Table 10 shows recent catches in the Convention Area, and the escapements to streams adjacent to it. Column 5 of Table 11 gives Canadian catches taken in the region from Area 11 south.

The period 1947-55 was characterized by intensified international rivalry for pink salmon in the southern region, and there was some tendency for catches to decrease in spite of the large increase in fishing effort, which suggests incipient overfishing (Table 10). In 1957 and 1959 the catch was down to approximately half the previous 5-year average, and in 1961 to about a ninth. In the latter year very severe limitation of fishing had to be introduced in order to avoid complete collapse of spawning populations. In 1963 there was a moderately good return of Fraser fish and an exceptionally good return to Puget Sound streams. In 1965 poor ocean survival created a crisis condition, with consequent restriction of the fishery. Preliminary returns in 1967 suggest at least reasonably good survival from the 1965 spawning.

An unfortunate feature of these recent events is that they have suppressed the build-up of pink salmon runs to tributaries of the Fraser River above the canyon. These had been destroyed at Hell's Gate in 1913, but since 1945 pinks have again been able to pass the canyon through the fishways completed in 1946. By 1957 the upriver pink salmon numbered 330,000 spawners, but subsequently they fluctuated and four generations later there were still only 358,000. The spawning grounds formerly used in the upper river region can accommodate some millions of fish, and according to one method of estimation they originally contained more than half of the Fraser system's spawning stock (INPFC Bull. 9, pp. 36-37).

PINK SALMON OF THE CENTRAL COAST OF BRITISH COLUMBIA

Catches from this region are shown in Table 11, under "Areas 5-10,30". Earlier data are given and discussed by Neave (1962), who has also plotted the trends for individual areas to 1959 (his figure 5). Whereas prior to 1945 the odd-year lines usually predominated in this region, since 1955 the even-numbered years have produced the biggest catches. The catch reached an all-time high of 20.6 million fish (81.7 million pounds) in 1962. In that year fishing restrictions were relaxed and equipment moved into the area in order to take about 75% of the total stock. The two subsequent cycle years have produced catches that are less spectacular but still good. In 1965 the odd-year line, after a relatively low escapement, experienced an unusual combination of drought followed by severe flooding, which resulted in extremely poor returns in 1967.

CHUM SALMON IN SOUTHERN BRITISH COLUMBIA

Recent trends in chum salmon catches are shown in Table 12 and Fig. 2. The first Canadian report (Anon. 1962a, p. 13-15) presented a graph of chum salmon landings and pointed out that an increase in rate of exploitation from 1945 to 1953 did not result in larger catches -- thus suggesting complete utilization. At the same time, many chum spawning grounds even then (1955) seemed underseeded, or at least contained substantially fewer spawners than previously. Since 1955 chum salmon catches and escapements both have decreased quite seriously in many areas, and a special study has been made of those tributary to the Strait of Georgia and Johnstone Strait.

In southern British Columbia most chums are taken in Johnstone Strait, where the catches include a mixture of stocks that spawn from that region south to the Fraser River and southern Vancouver Island. A smaller number of chums enter the Strait of Juan de Fuca and spawn mostly in Strait of Georgia tributaries.* Hence this whole region must be considered as a unit: that is, catches made in Statistical Areas 11-18, 28 and 29, and spawning streams that enter these areas (Fig. 16); it is unofficially called the "Johnstone Strait Study Area" (Palmer 1967).

In the study area adult chums are mainly of ages III (0.2) and IV (0.3). The fraction of the former in a given year-class has varied from 15% to 53%, average 41% (Palmer 1967). Figure 16 shows the catches from the study area and

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*Some chums are caught by United States fishermen in waters of the Strait of Juan de Fuca and Puget Sound; also, some chums spawn in United States streams and mix to some extent with Canadian stocks, and vice versa. However both the catches and the spawning escapements in northern Washington are small, and they are not considered here.

estimates of most of the spawning populations. The decline indicated has been caused in part, but only in part, by too great utilization of the returning stock.

Total returns from the spawnings of 1955 and subsequently have been computed from catches plus escapements, divided according to the age composition of the catch. There was exceptionally poor survival of the 1958 year-class, and to a less extent the 1961 year-class, which did not show up in quantity at either age 3 or age 4. The year 1958 was one of drought and high temperature in southern British Columbia, thus reproduction suffered.

Faced by declining escapements, increased restrictions have been put on fishing seasons, culminating in a practically complete closure in 1965 and 1966. The measures taken through 1963 were not sufficient to arrest the decline in stock, though the total escapement has declined less than the total stock; more recent closures could do little more than preserve the small runs for breeding stock.

A provisional estimate of spawning capacity in southern British Columbia is 2.5 million fish, whereas 1.6 million is the largest escapement estimated since 1949, and none since 1958 has exceeded 1 million. In addition, the distribution of spawners throughout this region in recent years has been far from even; in 1966 it varied from 8% to 52% of the estimated capacity of the spawning grounds in individual sectors of the coast.

Figure 17, considered uncritically, might suggest that production per spawner has declined really disastrously in the study area. Age composition data are not available to associate returns with spawnings earlier than the spawning of 1955. However, returns per spawner can be approximated back to 1949 (Table 13, Fig. 18). These don't suggest any really major change, though the average rate of return is about 36% less for 1956-62 than for 1949-55, in spite of the fact that populations are smaller. It is possible that many chum streams require a moderately large stock to maintain spawning grounds in good condition by their annual digging up of the redds, or depensatory predation on the fry may become critical for small stocks. There is obviously no suspicion of overseeding by the larger runs. In fact the rate of return is about the same for above-average spawnings as for those below average.

Our interpretation of these events is that during a period of reduced productivity the rate of utilization was not contracted quickly and severely enough to maintain (in many streams) the spawning reserve necessary to take advantage of a (probable) period of greater survival rate in years ahead. In anticipation of such a period, present restrictions on fishing are being maintained to conserve a reasonable breeding reserve.

If, however, the present reduced level of productivity were to be permanent and characteristic of all stock levels, the best course would be to continue fishing at a moderate rate (somewhat less than 50%) for the small catch that could be taken from stocks that would support a fishery, and let the remainder become commercially extinct.

It is worth noticing that Hoar (1951, p. 38) postulated three earlier periods of low chum salmon productivity in this area, suggested by an "index of return" computed from commercial landings. These periods were 1918-22, 1930-33 and 1943-47; or in terms of brood years approximately 1914-18, 1926-29 and 1939-43. Unfortunately the first two of these return periods partially coincide with periods of economic depression that may have tended to reduce catches, so the evidence for fluctuating productivity was not regarded as completely convincing. However the data take on new interest in the light of the present more protracted period of low productivity.

CHINOOK SALMON

The decline in Canadian chinook salmon landings (Table 1, Fig. 3) to below 10 million pounds during 1960-63 was a source of concern, but during the last three years they have risen again. Analysis of the stocks that contribute to the fishery is not yet sufficiently advanced to provide a good explanation for the observed changes. The chinook fishery is largely an international one, with United States and Canadian fishermen participating. The stocks fished are also produced in both countries. Increasing quantities of gear are being used, especially trolling gear; on the other hand net fishing, especially in rivers, has been curtailed somewhat because of restrictions dictated partly by the needs of other species.

On the whole the 22-year trend in chinook catches is slightly downward. Part of the reason for this can be associated with damage to various spawning streams by hydroelectric developments in the United States. Another factor is that with more intensive fishing, more fish are caught earlier in the year, long before they have completed their season's growth. Also, non-maturing chinooks are caught in some regions, and intensification of fishing effort on the feeding grounds leads to a smaller rate of survival to older ages. The average age and size of the fish caught in the troll fishery has declined greatly since the 1920's (Milne 1964).

The smallness of the year-to-year variations in chinook landings results mainly from the fact that these fish spawn mostly in rather large streams, hence their redds are less subject to drought or freezing than other species. The young of many chinook stocks stay in the river for only 3 or 4 months after emerging from the gravel, and leave before the time of summer low water. For stocks which remain in fresh water for a year the rivers characteristically have large summer flows. An additional factor contributing to stability is the fact that there are two age-groups present in important numbers in the chinook salmon catch, so the effects of unusually good or poor spawning success in a single year are averaged out.

COHO SALMON

Coho catches fluctuate more than chinooks, but less than other species. They have only one age in the catch in appreciable numbers (III's or 1.1). They require a full year of river life before migrating, and low flows can restrict their survival. The return in 1966 provided an exceptionally good coho catch, in fact the largest on record. Similarly the run in 1960 was the worst on record, and this is directly traceable to the hot dry summer of 1958 when many of the smaller nursery streams contracted greatly or even dried out.

To some extent Canadian coho stocks are captured by United States fishermen, and vice versa. The 22-year trend in Canadian landings is slightly upward, but this is small and non-significant. In view of the general increase in fishing effort, the failure of catches to increase importantly indicates that coho stocks are now being utilized to capacity.

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Table 1. Canadian landings of each species of salmon, 1958 to 1966 (1,000's of pounds). Based mainly on round weight, but chinook and coho figures are largely dressed weight, heads on. Source: BCCS. For earlier years see INPFC Bulletin 9, p. 7.

Year	Sockeye	Chinook	Coho	Pink	Chum	Total[a]
1958	74,011	12,412	22,356	33,750	38,104	181,321
1959	18,017	11,861	17,823	34,393	23,106	105,679
1960	15,470	9,021	12,846	16,914	20,313	75,154
1961	26,596	7,972	22,507	49,525	14,602	121,633
1962	20,077	7,911	24,146	93,214	18,046	163,908
1963	11,852	9,123	23,072	59,700	15,413	119,304
1964	22,922	12,093	28,588	36,447	23,914	124,199
1965	16,203	11,356	33,157	22,730	6,644	90,224
1966	25,653	13,615	35,030	72,894	15,354	162,854

[a] Includes a small quantity of steelhead trout.

Table 2. Trends in British Columbia salmon landings, 1951-66.

	21-year mean landings 1000's of lbs	Average annual increase or decrease 1000's of lbs	Relative annual increase or decrease %	Significance (5% level)
Sockeye	26,492	-380	-1.8	No
Chinook	11,906	-169	-1.5	Yes
Coho	23,236	+219	+0.8	No
Pink	46,218	+361	+0.9	No
Chum	36,604	-2468	-7.6	Yes
Total (includes steelhead)	145,079	-2485	-1.9	Yes

Table 3.

Commercial catches of Fraser River sockeye (Canada and the United States), subsistence catches, and estimated escapements (1000's of fish). Source: Annual Reports of INPFC.

	Commercial catch[a]	Subsistence catch[b]	Total catch	Escapement			Total (except jacks)
				Jacks[b]	Large[b]	Total	
1939	1,121	(45)	1,166	(7)	(127)	134	1,293
40	1,690	(45)	1,735	(28)	(439)	467	2,174
41	3,677	53	3,730	22	406	428	4,136
42	7,969	47	8,016	1	2,809	2,810	10,825
Total	14,457	(190)	14,647	(58)	(3,781)	3,839	18,428
1943	592	27	619	3	110	113	729
44	1,440	43	1,483	29	203	432	1,686
45	1,686	44	1,730	65	417	482	2,147
46	7,792	50	7,842	8	2,885	2,893	10,727
Total	11,510	164	11,674	105	3,615	3,920	15,289
1947	443	42	485	43	495	538	980
48	1,842	86	1,928	44	954	998	2,882
49	2,079	69	2,148	22	1,094	1,116	3,242
50	2,115	71	2,186	56	1,731	1,787	3,917
Total	6,479	268	6,747	165	4,274	4,439	11,021
1951	2,425	78	2,503	25	582	617	3,085
52	2,268	85	2,353	44	808	852	3,161
53	4,025	108	4,133	236	1,038	1,274	5,171
54	9,529	95	9,624	33	2,452	2,486	12,076
Total	18,247	366	18,613	338	4,880	5,229	23,493
1955	2,115	66	2,181	21	358	379	2,539
56	1,802	62	1,864	15	864	879	2,728
57	3,050	96	3,146	322	1,341	1,663	4,487
58	10,499[c]	82	10,581	45	3,771[c]	3,816[d]	14,352
Total	17,466	306	17,772	403	6,334[d]	6,737[d]	24,106
1959	3,393	65	3,458	22	935	947	4,393
60	2,454	86	2,540	7	613	620	3,153
61	2,735	138	3,873	90	1,163	1,253	5,036
62	1,595	135	1,730	49	1,574	1,623	3,304
Total	10,177	424	11,601	168	4,285	4,443	15,886
1963	2,001	190	2,191	24	1,576	1,599	3,767
64	1,023	145	1,168	42	390	431	1,558
65	2,065	121	2,186	78	767	845	2,953
66	2,687	154	2,841	121	1,798	1,919	4,639
Total	7,776	610	8,386	265	4,531	4,794	12,917

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Footnotes - Table 3

[a]Taken by Canada and the United States, in Convention waters only.

[b]Figures in brackets were estimated from the totals and trend of later years.

[c]An estimated additional 4.28 million Fraser sockeye were captured in waters north of the Convention area. (IPSFC Annual Report).

[d]Includes 1,006,000 sockeye diverted from Adams River into Shuswap Lake.

Table 4. Sockeye escapements ("jacks" omitted) to Fraser tributaries, 1960-66, and averages for each cycle starting with 1938 in the 1902 cycle (1000's of fish). Source: Annual Reports of the IPSFC. For earlier years see INPFC Bulletin 9, p. 100-101.

	1901 cycle				1902 cycle			
	1961	1965	Av.	%	1962	1966	Av.	%
Lower Fraser[a]	25.8	9.7	21.0	2.4	44.2	38.6	41.0	1.8
Harrison	47.4	26.8	35.5	4.0	24.6	52.5	43.2	1.9
Lillooet	31.7	8.2	42.3	4.8	26.4	20.1	40.8	1.8
Shuswap	5.8	10.1	8.6	1.0	1202.7	1311.7	1855.9	83.4
N. Thompson	7.8	6.7	5.6	0.6	7.7	6.3	5.6	0.3
Seton-Anderson	0.3	2.6	1.1	0.1	12.1	31.4	13.0	0.1
Chilcotin	39.2	35.3	135.7	15.3	78.4	210.0	71.8	3.2
Quesnel	302.3	364.6	146.9	16.5	1.1	1.8	0.9	<0.1
Nechako[b]	83.3	55.2	68.8	7.7	126.9	103.3	117.0	5.3
Early Stuart	201.0	23.0	176.9	19.9	25.4	10.8	23.5	1.1
Late Stuart	410.9	214.9	238.6	26.8	44.1	8.7	14.1	0.6
Bowron	7.5	2.7	9.1	1.0	6.3	2.5	7.6	0.3
Total escapement	1163.0	759.8	889.7	100.1	1599.9	1797.7	2224.1	99.9

[a]Includes Upper Pitt River, Cultus Lake and Widgeon Slough.

[b]This is the upper Nechako, above the Stuart River; called "Francois" in INPFC Bulletin 9.

Table 4 cont'd.

	1903 cycle			1904 cycle			
	1963	Av.	%	1960	1964	Av.	%
Lower Fraser	33.3	53.8	8.9	42.6	25.5	48.3	7.4
Harrison	36.7	23.0	3.8	29.3	7.4	33.3	5.1
Lillooet	48.9	40.3	6.6	35.9	48.9	49.3	7.6
Shuswap	228.1	129.4	21.3	6.1	4.4	9.0	1.4
North Thompson	9.3	6.8	1.1	5.5	5.7	8.5	1.3
Seton-Anderson	6.1	1.9	0.3	5.4	19.4	10.2	1.6
Chilcotin	1029.9	256.6	42.2	423.2	238.7	442.7	68.4
Quesnel	t	<0.1	<0.1	0.4	19.5	3.4	<0.1
Nechako	150.4	64.2	10.6	40.6	32.7	25.3	3.9
Early Stuart	4.6	12.5	2.1	14.4	2.4	12.0	1.9
Late Stuart	3.2	2.7	0.4	2.4	1.8	1.1	<0.1
Bowron	25.1	16.9	2.8	7.6	1.5	9.5	1.5
Total escapement	1575.6	607.3	100.1	613.4	407.9	647.3	100.1

Table 5. Four-year-total sockeye escapements to the Fraser River, including jacks (age 2+) (1000's of fish).

Source: Annual Reports of IPSFC.

	1939-1942	1943-1946	1947-1950	1951-1954	1955-1958	1959-1962	1963-1966
Cultus	202	68	62	69	75	108	52
Pitt	?	?	196	124	72	69	55
Harrison River	49+	33+	84	93	28	97	72
Harrison Lake[a]	60	77	85	92	93	41	52
Lillooet	189	296	387	229	140	181	248
Seton-Anderson	0	0	0	11	16	20	63
Early Shuswap[b]	3	3	38	64	109	120	115
Late Shuswap[c]	2612	2391	1504	2544	3659[d]	1355	1538
North Thompson	13	11	30	44	32	32	28
Chilcotin	616	595	824	854	1070	1043	1540
Quesnel	1	3	20	115	234	306	387
Nechako	59	280	362	371	307	335	342
Early Stuart	15	39	655	286	297	244	41
Late Stuart	5	26	151	364	559	439	229
Bowron	11	19	87	66	43	50	32
Total	3889+	3953+	4509	5228	6737[d]	4443	4894

[a]Includes Weaver Creek (= Morris Creek) as well as streams directly tributary to Harrison Lake.

[b]Includes Seymour, Anstey, Eagle, Middle Shuswap and Upper Adams Rivers, Scotch Creek early runs, and "miscellaneous streams".

[c]Includes the Lower Adams, Little and South Thompson Rivers.

[d]Includes 1,006,000 sockeye diverted from Adams River into Shuswap Lake in 1958.

Table 6. Rivers Inlet (Area 9) sockeye catches, 1956-1966 (1000's of fish). Source: BCCS. Data back to 1951 are in INPFC Bulletin No. 9, p. 51; estimates for earlier years are presented by Godfrey (1958, p. 333).

Year	Catch
1956	1,072
1957	374
1958	1,018
1959	440
1960	516
1961	843
1962	1,036
1963	438
1964	1,068
1965	645
1966	528
Total	7,978
Average (1956-66)	788.5

Table 7. Sockeye salmon catches from Statistical Area 4, and estimated subsistence catch and escapements in the Skeena River drainage (1000's of fish). Notice that important numbers of Skeena-produced sockeye may be caught in more distant waters, especially in Area 3 and in Alaska. Source: BCCS, and records of FRBC and CDFV.[e]

	Catch			Escapement[a]			Total (except jacks)
	Commercial	Subsistence	Total	Jacks[b]	Large[c]	Total	
1951	691	29	720	11	246	257	966[d]
1952	1295	46	1341	28	462	490	1803[d]
1953	659	38	697	28	765	793	1462
1954	572	31	603	10	588	598	1192
1955	157	17	174	31	109	140	283
1956	149	44	193	18	422	440	615
1957	282	55	337	50	505	555	842
1958	602	72	674	31	885	916	1559
1959	196	44	240	32	824	856	1064
1960	186	30	216	49	282	331	498
1961	895	48	943	28	975	1003	1918
1962	484	31	515	46	580	626	1095
1963	142	47	189	173	627	800	816
1964	766	40	806	60	881	941	1687
1965	294	39	333	64	652	716	985
1966	593	34	627	182	442	624	1069

[a] In some years the escapements of a few minor stocks were not estimated; these would number no more than 5000 fish in any one year.

[b] Includes jacks (3rd-year sockeye) in the Babine system only.

[c] Includes small numbers of jacks in areas other than Babine.

[d] In 1951 and in 1952 about two-thirds of the sockeye bound for the Babine system were blocked by a slide on the Babine River (Godfrey et al. 1954). Adding these fish in increases the estimate of total stock to 1,266,000 in 1951 and 2,503,000 in 1952.

[e] Commercial catches for earlier years are given by Godfrey (1958, p. 334).

Table 8. Sockeye spawners of life-history types 4_2 and 5_2 in the Skeena system, and the 4_2 and 5_2 stock (catch plus escapement) produced by each spawning (1000's of fish). Source: Shepard and Withler 1958, table I; Shepard et al. 1964, table I; and recent data compiled by H.W.D. Smith.

	Spawners	Resulting stock		
		4_2	5_2	Total
1908	1,124	1,138	520	1,658
1909	706	575	1,805	2,380
1910	1,506	613	1,308	1,921
1911	1,054	781	458	1,239
1912	765	513	381	894
1913	444	1,062	675	1,737
1914	979	1,775	1,882	3,657
1915	771	953	1,207	2,160
1916	393	273	177	450
1917	623	575	273	848
1918	970	1,603	849	2,452
1919	868	1,847	1,991	3,838
1920	546	782	841	1,623
1921	244	849	447	1,296
1922	625	1,093	394	1,487
1923	1,043	1,213	284	1,497
1924	1,098	389	469	858
1925	624	1,141	1,205	2,346

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Table 8 (cont'd).

	Spawners	Resulting stock		
		4_2	5_2	Total
1926	554	1,084	498	1,582
1927	549	920	424	1,344
1928	224	482	197	679
1929	565	336	341	677
1930	742	617	364	981
1931	510	518	371	889
1932	310	1,136	420	1,556
1933	173	420	220	640
1934	320	626	594	1,220
1935	310	850	333	1,183
1936	556	2,156	788	2,944
1937	340	763	439	1,202
1938	297	227	308	535
1939	598	273	1,045	1,318
1940	963	683	1,602	2,285
1941	572	486	697	1,183
1942	305	356	549	905
1943	272	171	306	477
1944	824	2,163	993	3,156
1945	940	218	577	795
1946	486	334	778	1,112
1947	307	397	514	911
1948	1,066	1,846	581	2,427
1949	480	759	604	1,363
1950	382	430	171	601

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Table 8 (cont'd).

	Spawners	Resulting stock		
		4_2	5_2	Total
1951	163	61	123	184
1952	158[a]	421	216	637
1953	700	524	825	1,349
1954	511	654	671	1,325
1955	87	252	127	379
1956	370	307	233	540
1957	448	1,569	610	2,179
1958	819	418	381	799
1959	799	357	1,126	1,483
1960	273	435	353	788
1961	954	526	456	982
1962	558	446		

[a]Estimate is for "effective spawners", corrected to account for the effects of the Babine River slide which blocked the river that year.

Table 9. Averages of annual estimates of spawning stocks in the minor sockeye areas of the Skeena system (1000's of fish). Some of these runs contain fish mainly of age 6_3, which are excluded from Table 8. Part of the increase shown for 1964-1966 may be due to the use of an improved method of estimation in some areas.

Period	Spawners
1944-48	186
1950-53	98
1954-58	70
1959-63	33
1964-66	58

Table 10. Commercial catches and escapements of pink salmon (1000's of fish) in the IPSFC Convention Area and escapements to streams adjacent to the Convention Area. Source: IPSFC Annual Reports, and IPSFC Bulletins No. 15 and 17.

	Catches			Escapements		
	Canada	USA	Total	Fraser	Other Canadian	USA
1945	1,280	5,459	6,739			
1947	3,491	8,802	12,293			
1949	3,190	6,235	9,425			
1951	2,886	5,086	7,972			
1953	4,142	4,951	9,094			
1955	4,129	4,686	8,815			
1957	2,635	2,777	5,412	2,424		
1959	2,313	2,428	4,740	1,078		
1961	545	509	1,054	1,094	733	808
1963	4,173	4,426	8,600	1,953	1,161	3,224
1965	592	558	1,151	1,191	160	812

Table 11. Annual pink salmon catches (1,000's of fish) in British Columbia, 1951-1966, by combinations of statistical areas. Source: BCCS.

	Areas					
	1 + 2	3 + 4	5-10,30	11-21 28,29	22-27	Total
1951	46	1,656	4,930	5,137	51	11,820
1952	2,761	1,742	3,800	2,722	210	11,235
1953	12	672	1,115	9,188	116	11,103
1954	1,515	1,302	2,052	403	171	5,444
1955	43	1,790	2,008	7,328	77	11,246
1956	404	1,422	4,410	924	193	7,352
1957	157	3,043	1,005	7,036	70	11,311
1958	1,209	1,583	2,709	1,378	30	6,908
1959	32	806	1,111	4,536	293	6,777
1960	413	239	3,091	350	5	4,098
1961	45	1,402	4,490	2,231	137	8,305
1962	1,073	874	20,622	760	100	23,429
1963	78	624	4,791	6,216	491	12,200
1964	348	1,322	7,078	863	18	9,628
1965	80	797	2,912	1,212	107	5,108
1966	1,843	1,705	10,106	3,454	153	17,261

Table 12. Annual chum salmon catches (1,000's of fish) in British Columbia, 1951-1966, by combinations of statistical areas. Source: BCCS.

	Areas					
	1 + 2	3 + 4	5-10,30	11-18, 28,29	19-27	Total
1951	611	348	1,616	2,782	445	5,804
1952	111	137	507	1,603	123	2,481
1953	142	230	1,226	2,409	663	4,670
1954	567	210	1,019	3,116	933	5,845
1955	83	98	429	717	241	1,568
1956	149	413	618	669	610	2,458
1957	86	262	1,101	559	404	2,412
1958	34	222	1,010	1,353	572	3,191
1959	20	202	205	1,254	337	2,018
1960	6	204	471	698	457	1,837
1961	71	130	458	369	190	1,217
1962	111	53	924	267	141	1,496
1963	145	56	806	357	99	1,463
1964	797	132	1,048	192	84	2,253
1965	180	49	343	36	25	633
1966	211	110	900	61	28	1,311

Table 13. Estimated escapements, returns (1,000's of fish) and number returning per spawner for chum salmon in the "Johnstone Strait Study Area". Source: Palmer (1967, p. 34) with the addition of the figures for 1949-54 in column 3, based on the average age composition of later catches.

	Spawning escapement	Return	Fish returning per spawner
1949	1,085	3,100	2.9
1950	1,419	3,800	2.7
1951	1,604	2,800	1.7
1952	1,271	1,300	1.0
1953	970	1,400	1.4
1954	1,039	1,800	1.7
1955	605	2,309	3.82
1956	489	1,432	2.93
1957	1,003	1,330	1.33
1958	1,149	283	0.25
1959	884	1,340	1.52
1960	581	1,005	1.73
1961	569	382	0.67
1962	649	865	1.33

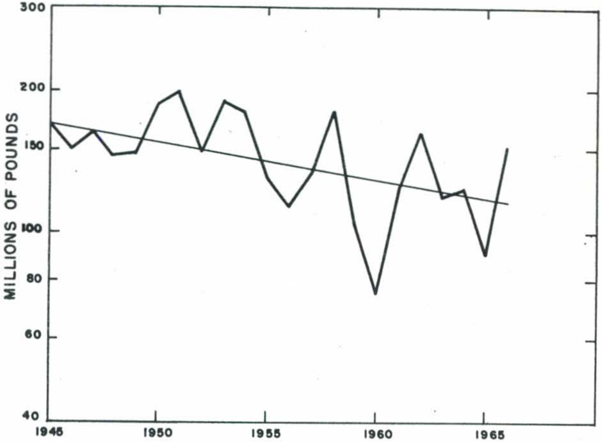

Fig. 1. Total landings of salmon in British Columbia, 1945-66 (logarithmic presentation).

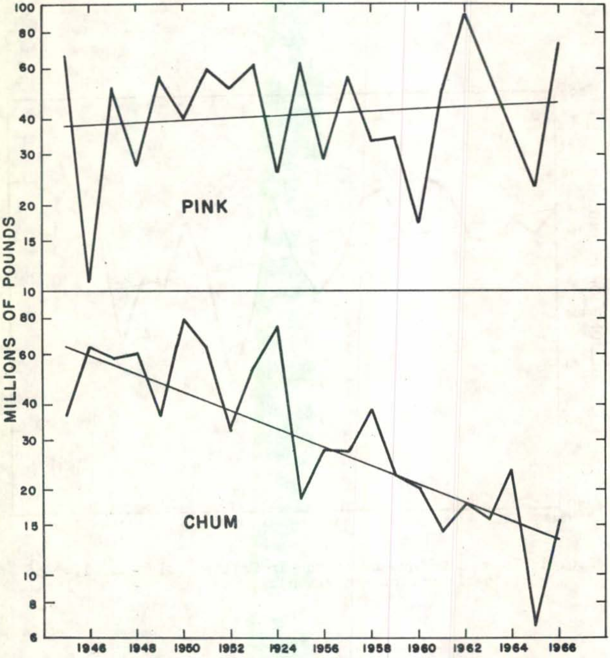

Fig. 2. British Columbia landings of pink and chum salmon (logarithmic presentation).

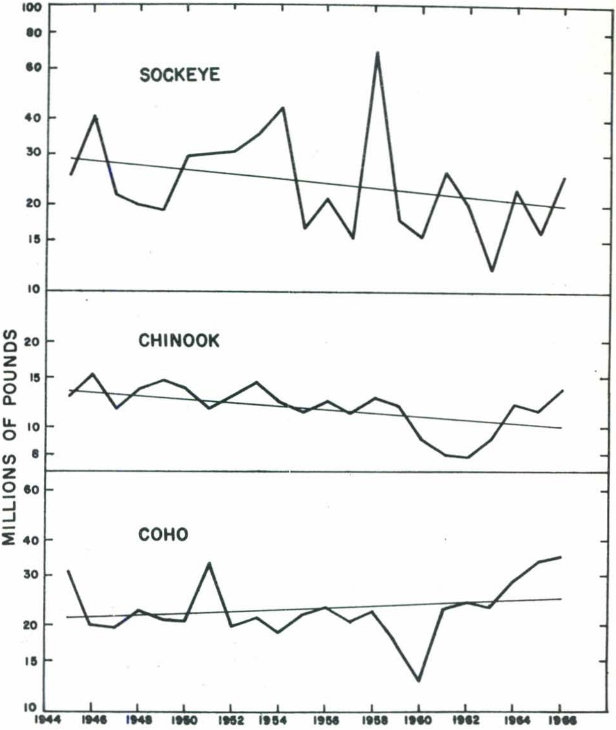

Fig. 3. British Columbia landings of sockeye, chinook and coho salmon (logarithmic presentation).

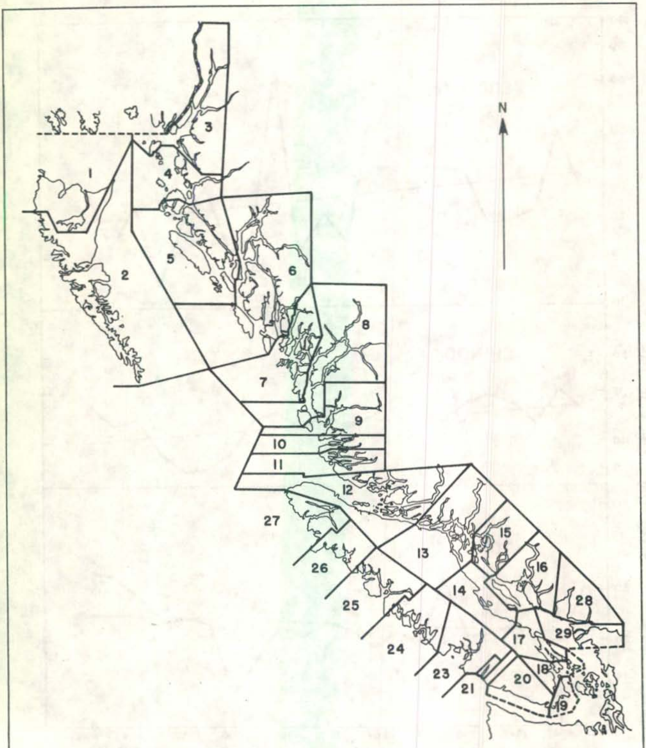

Fig. 4. Division of British Columbia coastal area into statistical areas.

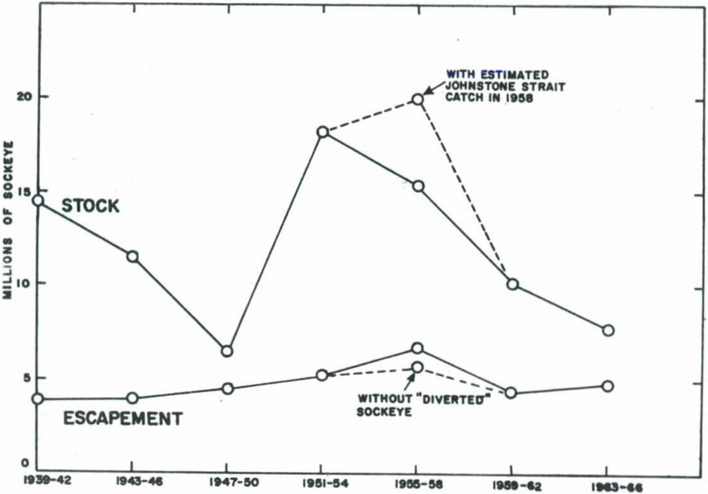

Fig. 5. Fraser River sockeye stock (Convention Area catch plus escapement), and escapements, by groups of four years. Data from Table 3.

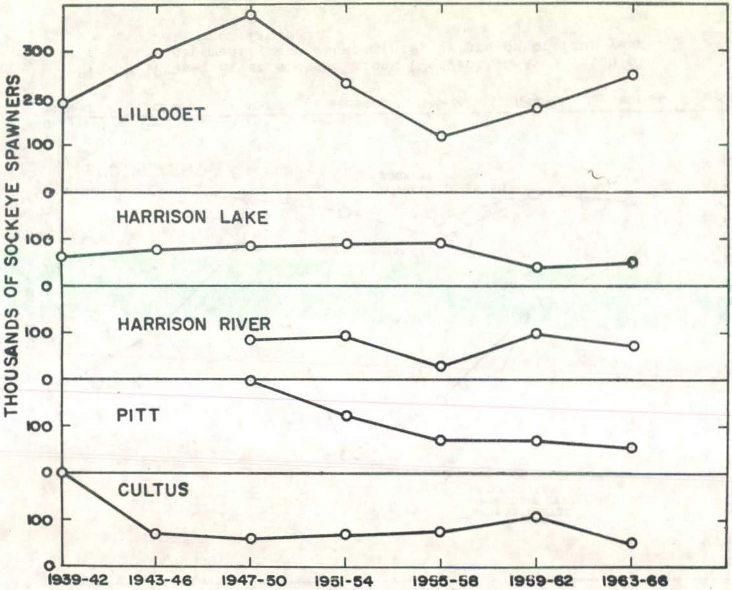

Fig. 6. Escapements of sockeye salmon to lower Fraser spawning areas. Data from Table 4.

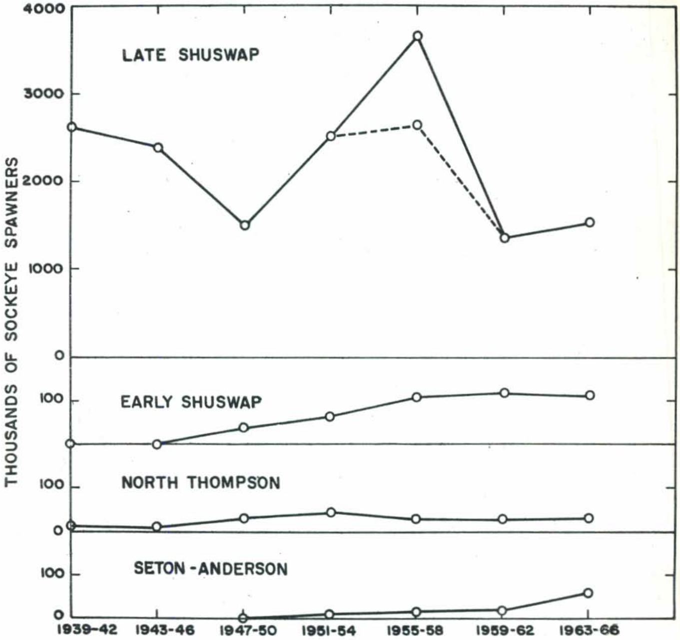

Fig. 7. Escapements of sockeye salmon to Seton-Anderson and Thompson River spawning areas.

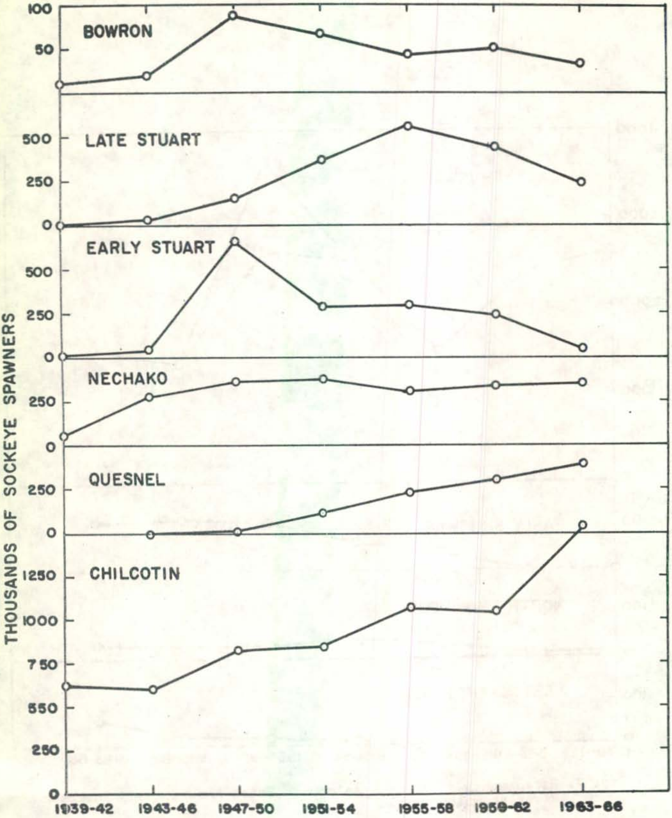

Fig. 8. Escapements of sockeye salmon to spawning areas of the Fraser River above Lillooet.

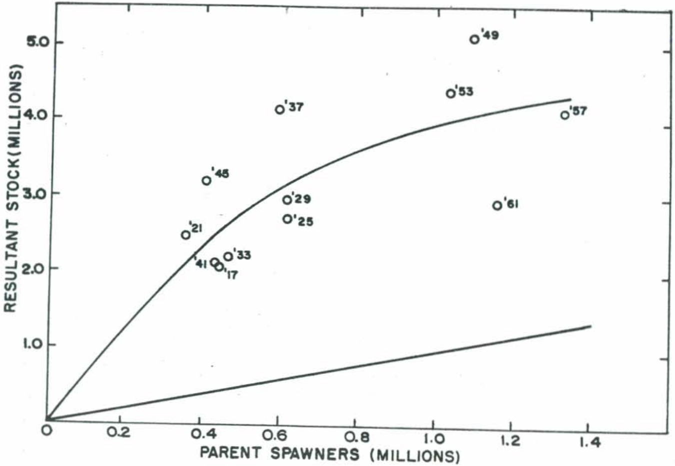

Fig. 9. Relation between escapement and return for Fraser sockeye of the 1901 cycle.

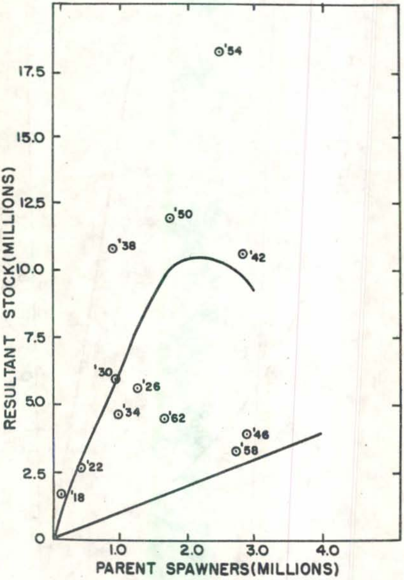

Fig. 10. Relation between escapement and return for Fraser sockeye of the 1902 cycle.

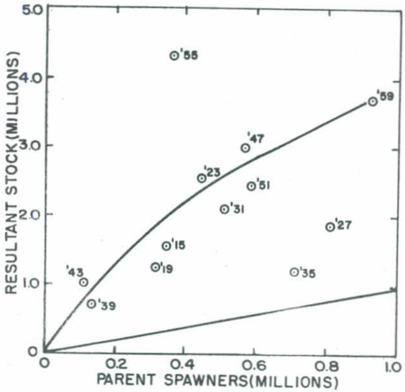

Fig. 11. Relation between escapement and return for Fraser sockeye of the 1903 cycle.

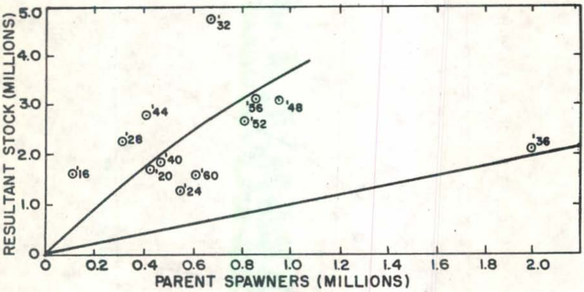

Fig. 12. Relation between escapement and return for Fraser sockeye of the 1904 cycle.

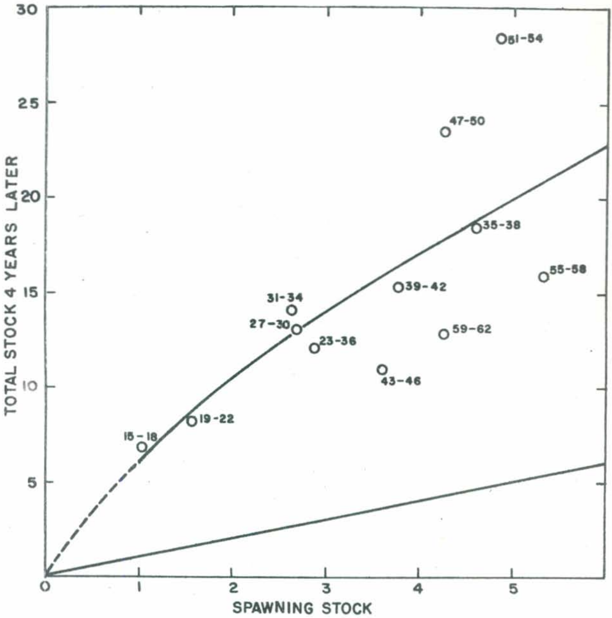

Fig. 13. Relation of estimates of spawners to total return for Fraser River sockeye, totalled by 4-year intervals, in millions of fish. Total stock includes commercial catch, subsistence catch (1939 onward) and spawning escapement (except jacks). Numbers by each point indicate the range of years included in the spawning escapement. (In 1958 the 1.0 million "diverted" sockeye are excluded from the spawning estimate, while the 4.3 million northern fish are included in the catch.) Source: Open circles are from Rounsefell (1949, p. 120); black dots are from Table 3.

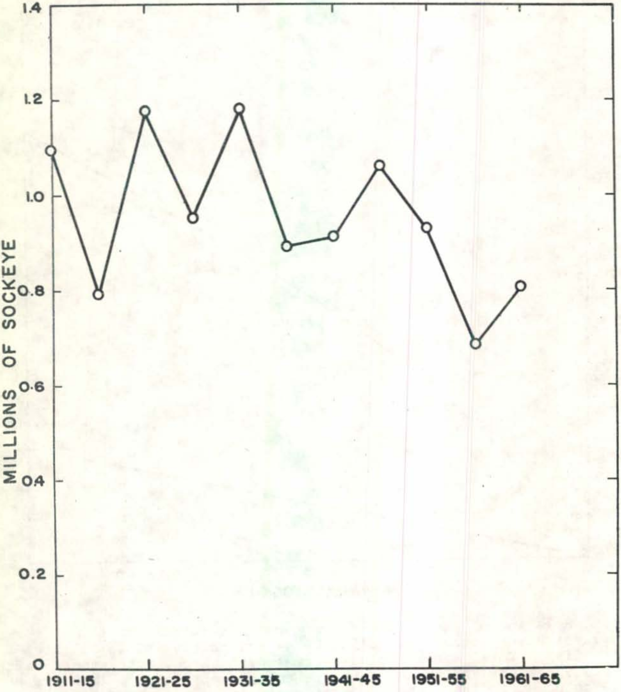

Fig. 14. Rivers Inlet sockeye catches, averaged by 5-year intervals, in numbers of fish. Data from Table 6 and references there cited.

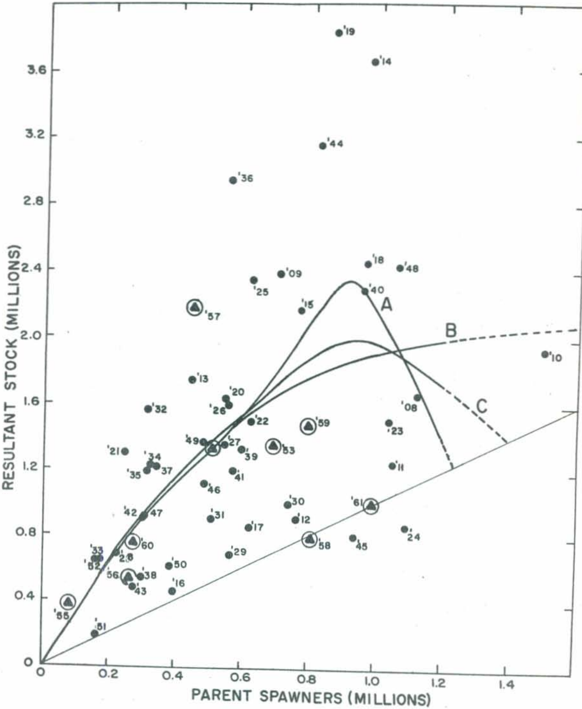

Fig. 15. Numbers of 4_2 and 5_2 Skeena sockeye (catch plus escapement) plotted against numbers of parent spawners. Figures in the body of the graph indicate brood years. Encircled points represent recent escapement-return values. Trend line A is the one drawn by Shepard and Withler (1958) on the basis of data through brood year 1952 (points without circles). Lines B and C are based on all the points shown, drawn to correspond as closely as possible to successive 5-year simple averages of the data when arranged in order of increasing numbers of spawners. Of the two latter lines, curve C is closer to the actual data; curve B represents a possible position if an abrupt decrease in reproduction above 0.9 million spawners be regarded as unlikely.

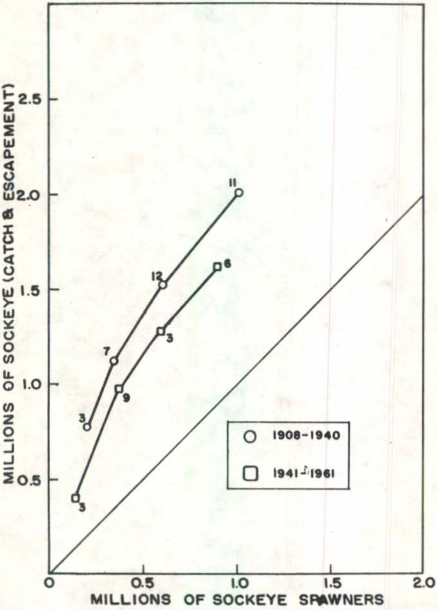

Fig. 16. Skeena sockeye 4_2 plus 5_2 escapements and returns for two periods, averaged by intervals 250,000 spawners broad.

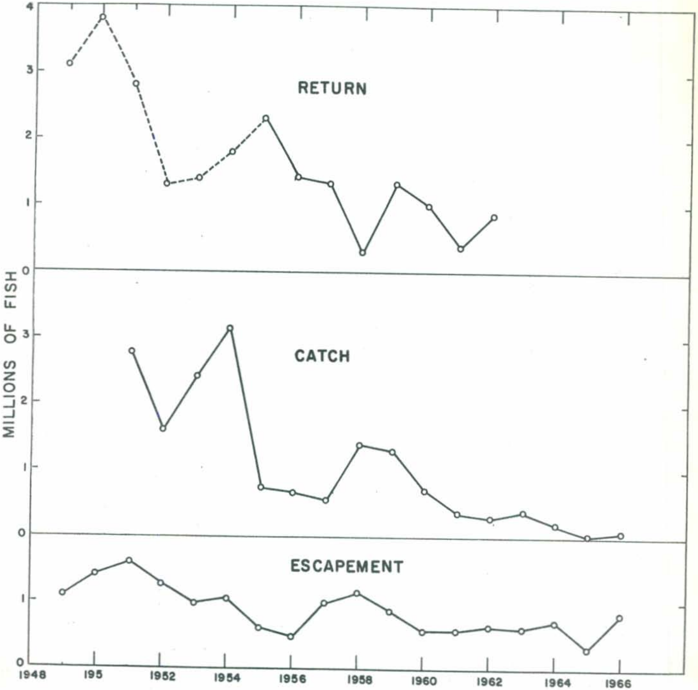

Fig. 17. Chum salmon catches and estimates of spawners through 1966, and the estimated return (catch plus estimated escapement) from the spawnings of 1949-62 for Areas 11-18, 28,29. The returns joined by broken lines are estimated from the average age composition of the catch. Data from Table 13.

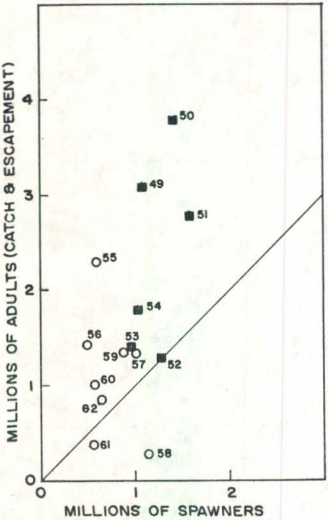

Fig. 18. Returns of chum salmon in the Johnstone Strait study area (catch plus estimated escapement) plotted against estimates of spawning stocks.

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